

Irsha Farwin

Full Stack Software Engineer | AI Engineer

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Professional Summary

Full Stack Software Engineer with 4+ years of experience building production web applications in React, Next.js, TypeScript, Node.js and Laravel/PHP, including a trilingual citizen-engagement platform in daily public use and a payment system that passed LKR 10 million in transactions within six months. Builds chat assistants on large language model (LLM) APIs, including a live portfolio assistant with prompt and context engineering, rate limits and streamed responses. Completing an MSc in Artificial Intelligence (AI), with a focus on reliable, grounded AI applications in production.

Skills

- **AI and Generative AI:** LLM APIs, Generative AI, Prompt and Context Engineering, Retrieval-Augmented Generation (RAG), Vector Databases, Streamed Responses, Rate Limiting, Python, Machine Learning, PyTorch, TensorFlow, Natural Language Processing (NLP), Computer Vision
- **Frontend:** React, Next.js, TypeScript, JavaScript, Tailwind CSS, Material UI, HTML, CSS
- **Backend:** Node.js, Laravel, PHP, RESTful APIs, GraphQL, API Design, Microservices
- **Databases:** MySQL, PostgreSQL, SQL, Firebase, Database Schema Design
- **DevOps and Cloud:** Docker, Continuous Integration and Continuous Delivery (CI/CD), Cloud Deployment, Git, GitHub, Postman
- **Practices:** Software Architecture, Agile, User Acceptance Testing (UAT), Monitoring, Debugging, Performance Optimization, Input Validation, Error Handling, Payment Gateway Integration, Third-Party API Integration

Professional Experience

Software Engineer

Sep 2025 - Present

Urban Development Authority (UDA) | Colombo, Sri Lanka (On-site)

- Owns the Mother Lanka Digital Wall end to end: a trilingual (Sinhala, Tamil, English) citizen-engagement platform in daily public use, with 6 of 7 modules live and the seventh in final Quality Assurance (QA).
- Contributes to UDA's first online payment system, the Financial Management Information System (FMIS), which passed LKR 10 million in transactions within six months of its February 2026 launch.
- Supports the move of 13,500+ housing residents and 1,400+ marketplace merchants from paper to digital payments through FMIS's integration with GovPay and payment gateways.
- Delivers modules for development plan and regulation publishing, public consultation and feedback, property bidding, development project oversight, and role-based user management with SMS notifications.
- Builds and integrates RESTful APIs connecting React, Next.js and TypeScript frontends to Laravel backend services and third-party systems.
- Improves monitoring, debugging and performance for government-scale traffic and compliance needs.
- **Technologies:** React, Next.js, Laravel, MySQL, RESTful APIs, JavaScript, TypeScript, Docker

Software Engineer (Contract)

Sep 2023 - Aug 2025

UnicornShift | Sydney, Australia (Remote)

- Delivered features end to end, from UI to API design and database optimization, for an AI-powered procurement and shift-management platform for civil infrastructure construction and maintenance.
- Built scalable full-stack features across the React web application and the Node.js and GraphQL API, connecting contractors and streamlining their operations.
- Collaborated remotely with a Sydney-based company across a 4.5 to 5.5 hour time difference.
- **Technologies:** React, Node.js, GraphQL, MySQL, PostgreSQL, Firebase, JavaScript, RESTful APIs

Software Engineer Intern

Jan 2022 - Jul 2022

Arthur C Clarke Institute for Modern Technologies (ACCIMT) | Colombo, Sri Lanka (On-site)

- Developed a Management Information System (MIS) for Buddhist and Pali University of Sri Lanka, handling trainee data and administrative workflows.

- Contributed to ACCIMT's internal MIS covering project management, procurement tracking, service agreements, automated attendance and financial management.
- Designed and maintained MySQL database schemas for efficient data storage and retrieval.
- Built frontend interfaces with Bootstrap and backend functionality with Laravel.
- **Technologies:** Laravel, PHP, Bootstrap, MySQL

Projects

- **AI Portfolio Assistant** | Personal project, live on portfolio website
 - Built a chat assistant that answers recruiters' questions about work experience and projects, using Next.js, React and TypeScript.
 - Integrated an LLM API (Groq, running the open-weight gpt-oss-120b model) with streamed responses.
 - Grounded answers in a system prompt built from portfolio data to control hallucination.
 - Implemented per-visitor rate limiting, a capped conversation history within token limits, input validation and error handling.
 - Delivered replies in English or Arabic, matching the visitor's language.
- **PathwayAI: University and Career Guidance Platform** | Researcher and Developer (MSc project)
AI platform that recommends university pathways to General Certificate of Education Advanced Level students from their results, interests and preferences. Python, Machine Learning, LLMs, Next.js, RESTful APIs.
- **LLM Chatbot for a Complaint Management System**
Integrated an LLM chatbot into a complaint management system.
- **UDA Mother Lanka Digital Platform** | Lead Engineer | motherlanka.uda.lk
Citizen-engagement platform for public feedback on development plans and regulations, with real-time stakeholder dialogue and access to urban planning resources. Next.js, Laravel.

Education

Master of Science (MSc) in Artificial Intelligence

Jan 2026 – Expected Jan 2028

Sri Lanka Institute of Information Technology (SLIIT) | Sri Lanka

Part-time, alongside full-time work

- Relevant modules: Machine Learning, Natural Language Processing, Neurocomputing and Neuroscience
- Frameworks used: PyTorch, TensorFlow

BSc (Hons) in Information Technology

Nov 2018 – 2023

University of Moratuwa | Moratuwa, Sri Lanka

- Final-year research: Automated Handwritten Bank Slip Digitalization (digital image processing)
- Relevant modules: Image Processing, Databases, Natural Language Processing, Programming

Research and Publications

- **AI-Powered University and Career Guidance Platform for Equitable Access in Sri Lanka**
First author. Presented at ICODE, Centre for Open and Distance Learning, University of Moratuwa, Aug 2026. Research on an AI platform giving Sri Lankan students more equitable access to university and career guidance; part of ongoing MSc research.

Certifications

- Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford Online (Coursera), May 2026
- Introduction to Front-End Development, Meta (Coursera), Dec 2025
- MySQL, Top 25%, TestDome, Dec 2022
- Responsive Web Design, freeCodeCamp, Dec 2020
- Introduction to SQL, DataCamp, May 2020

Languages

- English (Fluent), Tamil (Fluent), Sinhala (Conversational), Arabic (Reading)